NIAGARA 4 · REST API

glineRestApi v2

Feature Comparison

vs. Tridium built-in & third-party APIs

Shanghai Gline Net Co. · 2026

LANDSCAPE

The Options Today

Four ways to talk to a Niagara 4 station over HTTP

Tridium /rest/

Built-in REST endpoints. Requires SMA. Limited to station browsing.

oBIX

Built-in XML protocol. Strong for alarms & history. Verbose.

Third-party

Commercial connectors (e.g. ACTIVE-API). Per-station license cost.

glineRestApi v2

Single .jar. Batch ops, deep copy, BQL, built-in auth. No SMA.

COMPARISON

Feature Matrix

CAPABILITY	GLINERESTAPI V2	TRIDIUM /REST/	OBIX	THIRD-PARTY
Atomic batch (many ops, 1 request)	✓	—	—	—
Deep copy + link rebuild	✓	—	—	partial
Rename preserving links	✓	—	—	partial
Full component lifecycle	✓	read-only	partial	✓
BQL → structured JSON	✓	—	—	—
Built-in authentication	✓	✓	✓	✓
CIDR access allow-list	✓	—	—	partial
Alarms & history	—	partial	✓ (best)	✓
No SMA required	✓	✗	✓	varies
Single .jar deployment	✓	built-in	built-in	.jar

KEY DIFFERENCE #1

⚡ Atomic Batch Operations

One request. Many operations. All-or-nothing semantics.

glineRestApi v2

`POST /api/batch`

Send an array of operations — add, link, mod, read, invoke — and get an ordered array of results back.

Tridium /rest/ & oBIX

One endpoint call per object

Creating a device + 10 points + links means dozens of round trips, each with its own latency and failure handling.

KEY DIFFERENCE #2

📦 Deep Copy with Link Rebuild

Copy a folder — internal links are rebuilt automatically.

What it does

- Recursive component copy
- Internal links re-mapped src→dst
- External links preserved

Why it matters

- Template-based station cloning
- Repeat deployments in seconds
- No manual re-wiring

KEY DIFFERENCE #3

📊 BQL → Structured JSON NEW in v2

Query the station with BQL, get a table you can consume directly.

Request

`POST /api/batch`

```
{"OP":"bql","Q":"select name,slotPath,out from control:NumericWritable"}
```

Response

```
{"columns":["name","slotPath","out"], "rows":[...], "count":N}
```

- Feed reports, dashboards and AI pipelines directly
- No XML parsing, no extra transformation layer

KEY DIFFERENCE #4

🔐 Security Without a Proxy NEW in v2

Authentication and access control built into the module.

🔑 Basic Auth

Username + password. Configurable at runtime. Wrong or missing credentials are rejected.

🌐 CIDR Allow-list

Restrict access to trusted network segments. Multiple ranges supported.

💾 Persistent Config

Survives restarts. No external reverse proxy needed.

KEY DIFFERENCE #5

⚙️ Program Management NEW in v2

Create programs, write source, add slots and read status — all over HTTP.

📝

Write Source

`POST /api/program/write` — push program source to a live program.

+

Add Slot

`POST /api/program/addslot` — add input/output slots to a program.

📊

Read Status

`GET /api/program/status` — class, status, checksum, source length.

- v1 has none of this — program endpoints return `404` on v1
- Verified live on v2 (v2.0.3, port 8082)

RECOMMENDATION

When to Use What

USE CASE	RECOMMENDED
Batch point creation / template deployment	**glineRestApi v2 — `POST /api/batch`**
Station cloning with internal links	**glineRestApi v2 — deep copy**
Reporting & data extraction	**glineRestApi v2 — BQL to JSON**
AI agents configuring stations	**glineRestApi v2**
Alarm query & acknowledgement	oBIX (built-in)
Historical trend data	oBIX HistoryService

Bottom line: glineRestApi v2 + oBIX covers the full station management picture — lifecycle, queries, alarms and history — with no SMA and no third-party license.

VERIFIED

Test Results

40/40
Use cases passed

100%
Pass rate

40
Documented cases

29 functional + **11 access-control** cases · 11 batch operations · BQL · program management.
All requests verified with authentication enabled.

GET STARTED

Two Editions

v1 · niagaraRestApi

Available now — free

- No built-in auth
- 11 batch operations
- Port 8081

v2 · glineRestApi

On request

- Built-in auth + CIDR allow-list
- BQL to JSON
- Program management
- Port 8082

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